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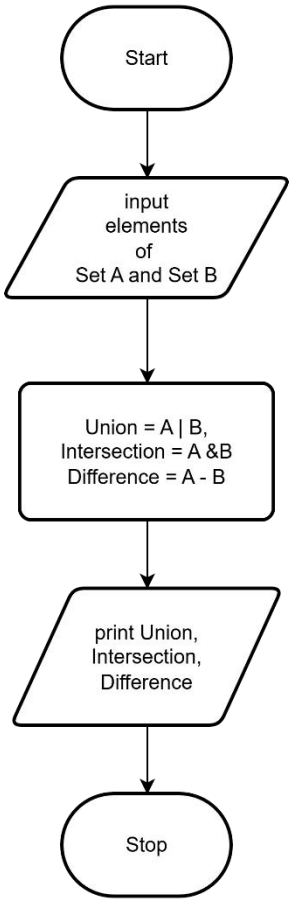
EXPERIMENT 4.1.1- SET OPERATION

ALGORITHM:

- **Start.**
- **Get Set A:** Ask the user for numbers, split them by spaces, and store them as a set of integers called set_a.
- **Get Set B:** Ask the user for numbers, split them by spaces, and store them as a set of integers called set_b.
- **Find Union:** Create a new set containing all unique numbers from both set_a and set_b.
- **Find Intersection:** Create a new set containing only the numbers that appear in both set_a and set_b.
- **Find Difference:** Create a new set containing numbers that are in set_a but **not** in set_b.
- **Display Results:** Print the results for Union, Intersection, and Difference clearly.
- **End.**

FLOWCHART:



CODE:

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4.1.1. Set Operations 11:34

Write a Python program to perform union, intersection and difference operations on $Set\ A$ and $Set\ B$.

Input Format:

- First Line prompts "Set A: " followed by space-separated list of integers for $Set\ A$.
- The second input prompts "Set B: " followed by space-separated list of integers for $Set\ B$.

Output Format:

- The first line prints "Union: " followed by the union of $Set\ A$ and $Set\ B$.
- The second line prints "Intersection: " followed by the intersection of $Set\ A$ and $Set\ B$.
- The third line prints "Difference: " followed by the difference of $Set\ A$ and $Set\ B$.

Sample Test Cases +

Explorer

setoperat... Submit

```python
set_a=set(map(int, input("Set A: ").split()))
set_b=set(map(int, input("Set B: ").split()))

union_set = set_a | set_b
intersection_set = set_a & set_b
difference_set = set_a - set_b

#Result
print("Union:", union_set)
print("Intersection:", intersection_set)
print("Difference:", difference_set)
```

Debugger

Average time 0.009 s 8.75 ms

Maximum time 0.014 s 14.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 14 ms Debug

Expected output Actual output

Terminal Test cases

Activate Windows
Go to Settings to activate Windows.